ModelIt!

Teacher's Guide

# Why Does Ice Cream Melt Faster on a Hot Day?

How Heat Energy Transfers from the Environment to Frozen Treats

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 05L2 |
| **NGSS Standard** | 5-PS1-1, 5-PS1-3 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

# Table of Contents

ModelIt!


1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

ModelIt!

**Performance Expectation: 5-PS1-1, 5-PS1-3**

Develop a model to describe that matter is made of particles too small to be seen. Make observations and measurements to identify materials based on their properties, including thermal conductivity and insulation.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop a model showing how air temperature and insulation control heat transfer and melting. |
| **Disciplinary Core Idea** | **PS1.A Structure and Properties of Matter** — Students model how thermal energy transfer causes changes in the state of matter from solid to liquid. |
| **Crosscutting Concept** | **Cause and Effect** — Students identify causal relationships between temperature difference, insulation, and melting speed. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

Two identical ice cream scoops are placed outside on a 95°F day. One is in a foam container and one is on a plate. Which melts first and why?

**CONSTRUCTED RESPONSE:**

Explain why ice cream melts faster on a hot day than on a cool day using the concepts of heat transfer and thermal energy from your model.

# 2 Scoring Rubric

ModelIt!

### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

ModelIt!

**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Heat transfer is one of the most fundamental concepts in physical science. Thermal energy always moves from warmer objects to cooler objects until they reach the same temperature — a state called thermal equilibrium. Ice cream melting is a perfect everyday example of this principle.

## What Is Heat Transfer?

Heat transfer is the movement of thermal energy from a warmer substance to a cooler substance. This happens through three mechanisms: conduction (direct contact), convection (moving fluids), and radiation (electromagnetic waves). Ice cream melting involves all three.

## Temperature Difference Drives the Rate

The greater the temperature difference between two objects, the faster heat transfers. On a 95°F day, heat rushes into 25°F ice cream much faster than on a 65°F day because the temperature gap is much larger.

## Insulation Slows Heat Transfer

Insulating materials like foam, wool, and trapped air have low thermal conductivity — they resist the flow of heat. A thermos uses vacuum-sealed walls and reflective linings to minimize all three types of heat transfer.

## State Change Requires Energy

Melting is a phase change that requires energy called the heat of fusion. The ice cream must absorb enough thermal energy to break the bonds holding its particles in a solid structure before it can become liquid.

# 4 The LEVER Framework

ModelIt!

### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**

Students identify Air Temperature, Container Insulation, Heat Transfer Rate, Melting Speed, and Ice Cream State as the five system components.

**E** **Establish Relationships**

Students determine that Air Temperature increases Heat Transfer Rate (+), Container Insulation decreases Heat Transfer Rate (-), Heat Transfer Rate increases Melting Speed (+), and Melting Speed changes Ice Cream State from frozen to melted (+).

**V** **Visualize & Model**

Students build the heat transfer model showing how external conditions (temperature and insulation) control internal responses (heat flow, melting, state change).

**E** **Evaluate Outcomes**

Students run the Insulated Container scenario and observe that high insulation dramatically slows melting even on a hot day, then compare with the Cool Day scenario to see the effect of temperature alone.

**R** **Revise & Extend**

Students add Wind Speed or Direct Sunlight to explore additional factors that affect heat transfer rate.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

ModelIt!

## Slide 1: Cover

2 min

**Say:** *"You know it happens — your ice cream melts faster on a hot day. But WHY?"*

**Watch For**

Ask students to share their worst ice cream melting story.

## Slide 2: Learning Objectives

2 min

**Say:** *"Today we model the invisible flow of heat energy."*

**Watch For**

Students read objectives and discuss what 'thermal energy' might mean.

## Slide 3: Big Question

2 min

**Say:** *"Everyone knows it happens, but can you explain the science behind it?"*

**Watch For**

Quick-write: students write their best explanation in 30 seconds.

## Slide 4: LEVER Framework

2 min

**Say:** *"We will model heat transfer as a system with inputs we control and outputs we observe."*

**Watch For**

Preview the five components students will work with.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Air Temperature and Container Insulation are external — we set them. The other three respond."*

**Watch For**

Students sort component cards into external and internal categories.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"Notice insulation has a NEGATIVE effect — it works AGAINST heat transfer. That is what makes it useful."*

**Watch For**

Guide students to draw all four relationship arrows in ModelIt.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's race: hot day vs insulated container. Which ice cream survives longer?"*

**Watch For**

Students run all three scenarios and record observations.

## Slide 8: Discoveries

5 min

**Say:** *"Heat always flows from hot to cold. Insulation is just a speed bump for that flow."*

**Watch For**

Class discussion: What real-world products use this principle?

## Slide 9: STEM Challenge

10 min

**Say:** *"Now design the ultimate ice cream keeper for the school fair."*

**Watch For**

Distribute materials; groups design, build, and test prototypes.

# 6 Answer Key & Expected Responses

ModelIt!

## Component Sorting

- **EXTERNAL:** Air Temperature, Container Insulation
- **INTERNAL:** Heat Transfer Rate, Melting Speed, Ice Cream State

## Relationships

- **Air Temperature to Heat Transfer Rate:** POSITIVE (+) — Higher air temperature creates a larger temperature difference, driving faster heat transfer.
- **Container Insulation to Heat Transfer Rate:** NEGATIVE (-) — Better insulation blocks heat from reaching the ice cream, slowing the transfer rate.
- **Heat Transfer Rate to Melting Speed:** POSITIVE (+) — Faster heat transfer delivers more thermal energy to the ice cream, speeding up melting.
- **Melting Speed to Ice Cream State:** POSITIVE (+) — Faster melting causes the ice cream to transition from frozen to slushy to liquid more quickly.

## Key Discoveries

- Heat always flows from warmer surroundings to cooler objects — never the other way
- Higher air temperature means a greater difference in thermal energy, so heat transfers faster
- Insulation slows heat transfer by blocking thermal energy from reaching the ice cream
- Melting speed depends on BOTH air temperature and insulation working together

# 7 Collaborative Learning Protocols

ModelIt!

### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

ModelIt!

### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

ModelIt!

### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Cold flows into the ice cream" | Cold is not a substance that flows. Heat (thermal energy) flows FROM the warm air INTO the cold ice cream. The ice cream warms up; the air does not send 'cold' anywhere. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Insulation adds cold to keep things frozen" | Insulation does not generate cold. It simply slows down heat transfer. A thermos keeps things cold by preventing warm air from reaching the contents, not by adding cold. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "All materials transfer heat at the same rate" | Different materials have very different thermal conductivities. Metal transfers heat rapidly while foam transfers it slowly. That is why a metal spoon in soup gets hot but a wooden spoon does not. |